



FORT STREET HIGH SCHOOL

2011

HIGHER SCHOOL CERTIFICATE COURSE
ASSESSMENT TASK 2

Mathematics Extension 1

TIME ALLOWED: 1½ HOURS

Name: _____

Teacher: _____

Class: _____

Outcomes Assessed	Questions	Marks
Manipulates algebraic expressions to solve problems involving inverse functions	1, 2, 4	
Applies appropriate techniques to solve problems involving parametric representations	3, 5	

Question	1	2	3	4	5	Total	%
Marks	/10	/11	/9	/12	/10	/52	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 (10 marks)

- a) Differentiate with respect to x : $y = \cos^{-1}(3x-1)$. 1
- b) Find the primitive function of
- i) $\frac{1}{1+3x^2}$ 1
- ii) $\frac{x+2}{x^2+4}$ 2
- c) Evaluate $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ 2
- d) Without evaluating the integral explain why $\int_{-\frac{1}{2}}^{\frac{1}{2}} \sin^{-1} x \, dx$ is equal to zero. 1
- e) Show that $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$. 3

Question 2 (11 marks)

- a) Consider the function $y = 2\cos^{-1} \frac{x}{3}$.
- i) Sketch the graph of this function clearly showing the domain and range. 2
- ii) Find the angle θ , that the tangent to the curve $y = 2\cos^{-1} \frac{x}{3}$ at $x=0$ makes with the positive direction of the x axis. Give your answer in degrees and minutes. 3
- iii) Find the area between this curve and the axes in the first quadrant. 3
- b) Sketch $y = \tan^{-1}(\sin 3x)$ for $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$, including any stationary points. 3

Question 3 (9 marks)

- a) The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ meets the x axis at T . The normal at P meets the y axis at N .
- i) Show that the equation of the normal at P is $x + py = 2ap + ap^3$. 1
- ii) Find the co-ordinates of M , the midpoint of TN . 2
- iii) Show that the locus of M is the parabola $x^2 = \frac{a}{2}(y - 2)$. 2
- b) The point $A(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$.
- i) Given that the normal at A passes through the point $R(-6a, 9a)$ show that $p^3 - 7p + 6 = 0$. 1
- ii) Hence, find the values of p on this parabola at which the normals intersect at R . 3

Question 4 (12 marks)

- a) Find the exact equation (in general form) of the tangent to the curve $y = \sin^{-1} \sqrt{x}$ at the point where $x = \frac{1}{2}$. 3
- b)
- i) Differentiate $x \cos^{-1} x - \sqrt{1 - x^2}$ with respect to x . 2
- ii) Hence evaluate $\int_0^1 \cos^{-1} x \, dx$. 2
- c) The area bounded by the curve $y = \frac{1}{\sqrt[4]{4 - x^2}}$, the x and y axes and the lines $x = 0$ and $x = \sqrt{3}$ is rotated about the x axis. Find the volume generated in exact form. 2
- d) Find showing all necessary working, the exact value of $\tan\left(2 \sin^{-1}\left(\frac{-1}{4}\right)\right)$. 3

Question 5 (10 marks)

- a) What is the Cartesian equation for the parametric equations $x = t - 2, y = t^2 - 4$? **1**
- b) Tangents from the point $P(x_0, y_0)$ touch the parabola $x^2 = 4y$ at Q and R .
- i) Prove that the midpoint T of QR is $\left(x_0, \frac{1}{2}(x_0)^2 - y_0\right)$. **5**
- ii) If P moves on the line $x - y = 1$, find the equation of the locus of T . **3**
- iii) Describe this locus geometrically. **1**

Question 1

a) $y = \cos^{-1}(3x-1)$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-(3x-1)^2}} \times 3$$

$$= \frac{-3}{\sqrt{1-(9x^2-6x+1)}}$$

$$= \frac{-3}{\sqrt{6x-9x^2}}$$

$$\frac{dy}{dx} = \frac{-3}{\sqrt{6x-9x^2}}$$

many forgot to multiply by ③. i.e. not using function of a function rule correctly.
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

b) i) $\int \frac{1}{1+3x^2} dx$

$$= \frac{1}{\sqrt{3}} \int \frac{x \sqrt{3}}{1+(\sqrt{3}x)^2} dx$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \sqrt{3}x + C$$

ii) $\int \frac{x+2}{x^2+4} dx$

$$= \int \frac{x}{x^2+4} dx + \int \frac{2}{x^2+4} dx$$

$$= \frac{1}{2} \ln(x^2+4) + \tan^{-1} \frac{x}{2} + C$$

c) $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx = \int \frac{1}{\sqrt{2^2-x^2}} dx$

$$= \left[\sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}}$$

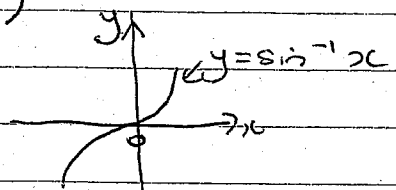
$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6}$$

mostly well done.

Q1 d)



As $y = \sin^{-1} x$ is an odd fn it has pt. symmetry. ie. the area above the x

many did not explain clearly how the area below and above the x -axis cancels each other out.

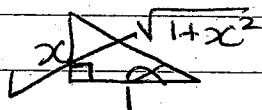
equals the area below the axis & $\therefore$ cancels each other out.

e) Show that $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$

mostly well done

Let $\tan^{-1} x = \alpha$

$$\therefore x = \tan \alpha$$



$$\therefore \cos \alpha = \frac{1}{\sqrt{1+x^2}} \quad (\text{using rt. A})$$

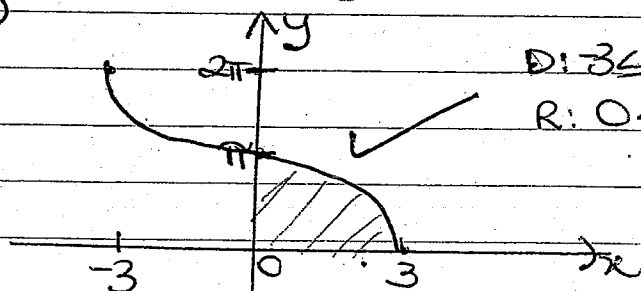
$$\therefore \alpha = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$

$$\text{but } \alpha = \tan^{-1} x$$

$$\therefore \tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$

QUESTION 2

a) i) $y = 2 \cos^{-1} \frac{x}{3}$



$$\begin{aligned} \text{D: } -3 \leq x \leq 3 \\ \text{R: } 0 \leq y \leq 2\pi \end{aligned}$$

mostly well done

ii) $f(x) = 2 \cos^{-1} \frac{x}{3}$
 $f'(x) = 2 \cdot \frac{-1}{\sqrt{9-x^2}}$

$$= -\frac{2}{\sqrt{9-x^2}}$$

$$f'(0) = -\frac{2}{3}$$

$$\begin{aligned} \therefore \tan \theta &= -\frac{2}{3} \\ \theta &= 146^\circ 19' \\ &(\text{nearest min}) \end{aligned}$$

poorly attempted
many did not make the connections between $f'(x) = m = \tan \theta$

Q2 (cont'd)

2a) iii) $A = \int 2 \cos^{-1} \frac{x}{3} dx.$

$$y = 2 \cos^{-1} \frac{x}{3}$$

$$\frac{y}{2} = \cos^{-1} \frac{x}{3}$$

$$3 \cos \frac{y}{2} = x$$

$$A = 3 \int_0^{\pi} \cos \frac{y}{2} dy \quad \checkmark$$

$$= 3 \times 2 \left[\sin \frac{y}{2} \right]_0^{\pi} \quad \checkmark$$

$$= 6 \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$= 6 (1 - 0) \quad \checkmark$$

$$A = 6 \text{ u}^2.$$

some tried to integrate this directly rather than finding the area bounded by the curve and y-axis

b) $y = \tan^{-1}(\sin 3x)$

$$\frac{dy}{dx} = \frac{1}{1 + \sin^2 3x} \cdot 3 \cos 3x$$

for statnary pts $\frac{dy}{dx} = 0.$

$$\therefore 3 \cos 3x = 0$$

$$\cos 3x = 0$$

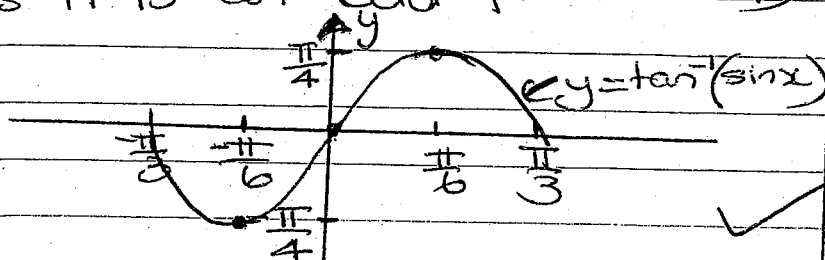
$$3x = -\frac{\pi}{2}, \frac{\pi}{2}$$

$$x = -\frac{\pi}{6}, \frac{\pi}{6} \quad \checkmark$$

when $x = \frac{\pi}{6}$:

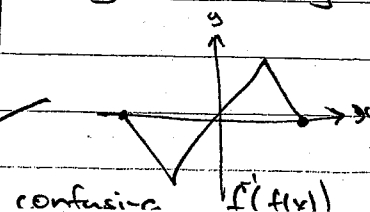
x	$\frac{\pi}{6}$	0	$\frac{\pi}{6} +$
y	+	0	-

$\therefore$ maxm statnary pt at $(\frac{\pi}{6}, \frac{\pi}{4})$
 also minm statnary pt at $(-\frac{\pi}{6}, -\frac{\pi}{4})$
 as it is an odd fn $(\frac{\pi}{6}, \frac{\pi}{4})$



many did not confirm/test the nature of the stationary points

many incorrectly had



confusion

QUESTION 3.

a) $P(2ap, ap^2) \quad x^2 = 4ay$

i) $y = \frac{x^2}{4a}$

$y' = \frac{x}{2a}$

at $x = 2ap \quad y' = p$

$\therefore$ grad. normal $= -\frac{1}{p}$
 eqn of normal: $y - ap^2 = -\frac{1}{p}(x - 2ap)$
 $py - ap^3 = -x + 2ap$
 $\therefore x + py = 2ap + ap^3$
 (as reqd)

well done.

ii) Eqn of tangent is:

$y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$

at T $y = 0 \Rightarrow px = ap^2$

$x = ap$

$\therefore T(ap, 0)$
 $N(0, 2a + ap^2)$

$\therefore M\left(\frac{ap}{2}, \frac{2a + ap^2}{2}\right)$

$\Rightarrow M\left(\frac{ap}{2}, a + \frac{1}{2}ap^2\right)$

Students need to find T and N to get one mark.

Some mixed up when to let $x=0$ & $y=0$, which resulted in error carried for pt M & made it much harder.

iii) $x = \frac{ap}{2}$

$\therefore p = \frac{2x}{a}$

$y = a + \frac{1}{2}ap^2$

$2y = 2a + a\left(\frac{2x}{a}\right)^2$

$= 2a + 4x^2/a$

$2ay = 2a^2 + 4x^2$

$4x^2 = 2ay - 2a^2$

$x^2 = \frac{ay - a^2}{2}$

$x^2 = \frac{a}{2}(y - a)$

is the locus of

many ignored the 'typo' & got correct answer.

Others had correct answer & just changed it to a^2 for convenience.

- Marked very leniently.

- Others totally ...

Q3 b) $A(2ap, ap^2) \quad x^2 = 4ay$

i) normal is: $x + py = 2ap + ap^3$

thru $(-6a, 9a)$: $-6a + 9ap = 2ap + ap^3$
 $-6 + 7p = p^3$

$\therefore p^3 - 7p + 6 = 0$ (as reqd)

No need to derive formula again, just state it.

ii) Let $f(p) = p^3 - 7p + 6$
 $f(1) = 0$

$\therefore (p-1)$ is a factor ✓

$$\begin{array}{r} p^2 + p - 6 \\ (p-1) \overline{) p^3 - 7p + 6} \\ \underline{p^3 - p^2 + 6p - 6} \\ p^2 - 7p + 6 \\ \underline{p^2 - p} \\ -6p + 6 \\ \underline{-6p + 6} \\ 0 \end{array}$$

$\therefore p^2 + p - 6 = (p-1)(p^2 + p - 6)$
 $= (p-1)(p-2)(p+3)$ ✓

$\therefore p = 1, 2, -3$

many had trouble finding any factor first & hence made it hard to complete.

QUESTION 4:

a) $y = \sin^{-1} \sqrt{x}$

$y' = \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{2} x^{-\frac{1}{2}}$

$= \frac{1}{2\sqrt{x(1-x)}}$

$= \frac{1}{2\sqrt{x-x^2}}$ ✓

when $x = \frac{1}{2}$, $y' = \frac{1}{2\sqrt{\frac{1}{4}}}$

$\therefore m = 1$

when $x = \frac{1}{2}$, $y = \sin^{-1} \frac{1}{\sqrt{2}}$

$y = \frac{\pi}{4} \Rightarrow \left(\frac{1}{2}, \frac{\pi}{4}\right)$

$\therefore y - \frac{\pi}{4} = 1 \left(x - \frac{1}{2}\right)$

$4y - \pi = 4x - 2$

$4x - 4y - 2 + \pi = 0$ ✓
 is the tangent.

• many did not use the chain rule correctly here (yet did so in pt(b))

both ✓

• general form has no fractions in it!

4(b)i) let $y = \underbrace{x}_{u} \cos^{-1} \underbrace{x}_{v} - \sqrt{1-x^2}$

• many had problems with signs, so did not eliminate terms.

$$y' = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} - \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot (-2x)$$

$$= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$$

$$y' = \cos^{-1} x \quad \checkmark$$

ii) $\int \cos^{-1} x \, dx$

$$= \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_0^1 \quad \checkmark$$

$$= (\cos^{-1} 1 - 0) - (0 - 1) \leftarrow$$

$$= 0 + 1$$

$$= 1 \quad \checkmark$$

• mostly well done - many got -1 as an answer because they didn't use brackets appropriately!

(c) $V = \pi \int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} \, dx$

$$= \pi \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} \quad \checkmark$$

$$= \pi \left(\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \right)$$

$$= \pi \left(\frac{\pi}{3} - 0 \right)$$

$$V = \frac{\pi^2}{3} \quad \checkmark$$

• some strange variants on this standard integration - use the 'Tables of Standard Integrals' provided!

(d) $\tan \left(2 \sin^{-1} \left(-\frac{1}{4} \right) \right)$

$$= -\tan \left(2 \sin^{-1} \frac{1}{4} \right)$$

let $\sin^{-1} \frac{1}{4} = x$

$$\therefore \sin x = \frac{1}{4} \quad \leftarrow \text{triangle with hypotenuse 4, opposite 1}$$

now $\tan \left(2 \sin^{-1} \left(-\frac{1}{4} \right) \right) = -\tan 2x$

$$= \frac{-2 \tan x}{1 - \tan^2 x}$$

$$= \frac{-2 \times \frac{1}{\sqrt{15}}}{1 - \frac{1}{15}} \quad \checkmark$$

$$= \frac{-2}{\sqrt{15}} \times \frac{15}{14}$$

$$= \frac{-15}{7\sqrt{15}} \times \frac{\sqrt{15}}{\sqrt{15}}$$

$$\tan \left(2 \sin^{-1} \left(-\frac{1}{4} \right) \right) = -\frac{\sqrt{15}}{7} \quad \checkmark$$

• many errors here, usually related to not understanding the restricted domain/range when dealing with inverse trig functions!

• Also simple trig identity errors with

$$\sin(-x) = -\sin x$$

$$\tan(-x) = -\tan x$$

• many had $1 + \tan^2 x$ for their denominator!

Question 5:

a) $x = t - 2$ $y = t^2 - 4$
 $t = x + 2$ $y = (x + 2)^2 - 4$
 $\quad\quad\quad = x^2 + 4x + 4 - 4$
 $\quad\quad\quad \boxed{y = x^2 + 4x} \quad \checkmark$

To obtain mark students had to have in simplified form.

b) $x^2 = 4ay$
 $\Rightarrow x^2 = 4y \quad \therefore a = 1$
 Let $R(2ar, ar^2) \Rightarrow R(2r, r^2)$
 $Q(2aq, aq^2) \Rightarrow Q(2q, q^2)$

$$m_{RQ} = \frac{r^2 - q^2}{2r - 2q}$$

$$= \frac{(r - q)(r + q)}{2(r - q)}$$

$$m_{RQ} = \frac{r + q}{2} \quad \checkmark$$

eqn of QR:

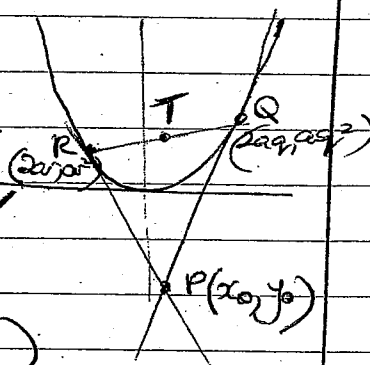
$$y - q^2 = \frac{r + q}{2}(x - 2q)$$

$$y - q^2 = \frac{(r + q)}{2}x - q(r + q)$$

$$y - q^2 = \frac{(r + q)}{2}x - q^2 - rq$$

$$2y + 2qr = (r + q)x$$

$$x(r + q) = 2(y + qr) \quad \checkmark \quad \text{--- (1)}$$



Students attempted various ways of doing this and often got into a real mess

Now eqn of chord ^{of contact} from external pt

is $xx_0 = 2a \left(\frac{y + y_0}{2} \right)$ $\checkmark$ (2)

comparing (1) & (2)

$$x_0 = r + q \quad \text{and} \quad y_0 = qr \quad \checkmark$$

New midpt of QR: $\left(qr, \frac{r^2 + q^2}{2} \right)$

$$= \left(qr, \frac{(r + q)^2 - 2qr}{2} \right) \quad \checkmark$$

$$= \left(x_0, \frac{1}{2}(x_0)^2 - y_0 \right) \text{ as req'd.}$$

ii) If P moves on $x-y=1$
 $y=x-1$
 then $y_0 = x_0 - 1$

Locus of T: $x = x_0$; $y = \frac{1}{2}(x_0)^2 - y_0$
 $= \frac{1}{2}(x_0)^2 - (x_0 - 1)$ ✓
 $= \frac{1}{2}x^2 - (x - 1)$
 $= \frac{1}{2}x^2 - x + 1$
 $= \frac{1}{2}(x^2 - 2x + 1) + \frac{1}{2}$ ✓

$y = \frac{1}{2}(x-1)^2 + \frac{1}{2}$
 $y - \frac{1}{2} = \frac{1}{2}(x-1)^2$
 $2y - 1 = (x-1)^2$

∴ Locus is $(x-1)^2 = 2(y - \frac{1}{2})$ ✓

iii) this is a parabola (concave up) Need to give
 $v(1, \frac{1}{2})$ focal length = $\frac{1}{2}$ unit. more
 information
 other than it is
 a parabola